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FEDERAL SPECIFICATION

LACES, FOOTWEAR, SPUN NYLON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — This specification covers one type of spun-nylon laces for use as closures for various types of footwear (see 7.2.6.3).

1.2 Classification.

1.2.1 *Classes and lengths.*—Laces shall be of the following classes and lengths, as specified:

Class 1.—7/32-inch width.
Lengths.—25, 27, 28½, 30, 38, 40, 50, 52, 54, and 84 inches.
Class 2.—1 9/32-inch width.
Lengths.—30, 40, 44, 52, 60, 64, 72, and 80 inches.

1.2.2 *Grade.*—Laces shall be furnished in the grade known commercially as firsts.

1.2.3 *Color.*—Laces shall be of the color as specified (see 6.1).

2. APPLICABLE SPECIFICATION

2.1 The following Federal specification, of the issue in effect on date of invitation for bids, forms a part of this specification:

CCC-T-191—Textile Test Methods.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by check, money order, cash or Government Printing Office coupons, to the General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S. W., Washington, 25, D. C. This office will also honor deposit account numbers issued by the Government Printing Office. Single copies of product specifications required for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C. Prices of additional copies may be obtained from the GSA Regional Offices.)

2.2 Specifications and other publications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Material.

3.1.1 *Nylon.* — The nylon shall be first quality, cut staple and of such staple length and denier to produce laces conforming to the specification. The use of regenerated nylon or nylon waste is prohibited.

3.1.2 *Yarns.* — The nylon staple shall be evenly spun and twisted into a two-ply yarn (see 6.4).

3.1.3 *Tips.*—The tips shall be made from cellulose acetate or cellulose-acetate butyrate having a thickness of not less than 0.010 inch. The tips shall be transparent and colorless or approximately the same color as the lace.

3.2 Color.—The yarn shall be dyed with applicable chromiferous (metallized) chrome, acid, direct dyes or combinations thereof. The yarns shall be well soaped and washed

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after dyeing. The use of sulphur dyes is prohibited.

3.2.1 *Color matching.* — The color shall match the specified shade (see 3.2 and 7.2.2) under natural (north sky) daylight or artificial daylight having a color temperature of 7,500° K. and shall be a good approximation to the specified shade under incandescent lamplight at 2,800° K.

3.2.2 *Colorfastness.*—The dyed laces shall show good color fastness to crocking, perspiration, and light when tested as specified in 4.3.

3.3 **Braid.**—The braid for the laces shall be tubular without a core, pressed flat, and shall conform to the following:

3.3.1 *Weave.*—The weave shall be a plain weave (over two and under two).

3.3.2 *Physical requirements.* — The braid shall conform to the physical requirements shown in table I.

3.4 **Tipping.**—Both ends of the braid shall be provided with a firmly attached and secured tip made from the material as specified in 3.1.3. Tips shall be $\frac{9}{16} \pm \frac{1}{16}$ inch long.

3.4.1 *Gripping strength.* — The tips shall not be dislodged from their position on the laces when tested as specified in 4.3.4.

3.5 **Width.**—Class 1 laces shall be $\frac{7}{32}$ inch in width and class 2 laces shall be $\frac{10}{32}$ inch in width. The width shall be determined as described in 4.3.1.

3.6 **Length.**—Laces shall be furnished in lengths, including tips, as specified (see 6.1). Class 1 laces shall be 25, 27, 28½, 30, 38, 40, 50, 52, 54, and 84 inches long. Class 2 laces shall be 30, 40, 44, 52, 60, 64, 72, and 80 inches long. The laces shall be cut so as to insure the specified length when tested as specified in 4.3.2. A tolerance of plus 5 percent in length will be permitted.

3.7 **Workmanship.** — The finished laces shall be clean, evenly braided, and free from any defects which may affect their appearance or serviceability.

TABLE I.—*Physical requirements*

Class	Width*	Ends per carrier	Carriers, minimum	Picks per inch, minimum	Breaking strength, minimum
	Inch				*Pounds*
1	7/32	1	44	38	100
2	10/32	2	44	26	175

* A tolerance of ±1/32 inch will be permitted.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling.**—Samples for inspection and test shall be taken at random in an amount equal to one percent of each lot, but not less than 6 nor more than 20 laces. For the purpose of this specification, a lot shall be considered as consisting of that quantity of laces of the same class, color, and length offered for inspection at any one time or being within a single shipment.

4.2 **Inspection.**—Inspection shall be made at the point of delivery or at the factory at the discretion of the purchasing agent.

4.3 **Tests.**—Tests shall be conducted in accordance with the following methods described in Federal Specification CCC–T–191 and as hereinafter specified:

Tests	Method No.
Standard condition	Section 4
Identification of nylon	1530
Crocking resistance; yarn, thread, cordage	4650
Colorfastness to light; yarn, thread, cordage; accelerated method (Fade-ometer)	4660
Colorfastness to perspiration of cloth	4680 or 4682

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4.3.1 *Width.* — The width shall be determined at three points at least 9 inches apart by a steel rule calibrated in thirty-seconds of an inch and the results averaged. The width shall be measured flat with the test lace under no tension.

4.3.2 *Length.*—The finished-length measurement shall be taken from end to end, including the tip, when the finished lace is suspended vertically under a tension of 3 ounces for class 1 and 6 ounces for class 2 laces.

4.3.3 *Breaking strength.* — The breaking strength of the laces shall be determined as described in Federal Specification CCC–T–191, method 4102, except the clamps shall be not less than 2 inches in diameter and the distance between the center of the clamps at the start of the test shall be 6 ± 1/16 inches.

4.3.4 *Tests for tips.*—A rigid metal plate having a narrow slot is required. The slot shall be not less than 4 inches long, 1/4 inch wide at one end and 1/32 inch wide at the other end. The slot shall be smoothly finished. It is not required that the plate be of a single piece; the slot may be formed by a clamped or bolted assembly of two metal bars or plates properly spaced apart. The plate shall be supported in a horizontal position. The lace shall be passed upward, tip first, through the wide end of the slot until the tip clears the plate, and then moved horizontally toward the narrow end of the slot until the end of the tip rests on top of the metal. To the hanging end of the lace, about 10 inches from the tip, a 20 pound weight shall be gently attached.

5. PREPARATION FOR DELIVERY

5.1 **Packaging.**—Commercial packages will be acceptable.

5.2 **Packing.** — The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 **Marking.**

5.3.1 *Issue packages.* — Issue packages shall be marked with the class, color, length, and name or trade-mark of the contractor.

5.3.2 *Shipping containers.*—Shipping containers shall be marked with the name of the material, the class, length, and color, as defined by the contract or order under which the shipment is made, and the quantity contained therein, the number of the contract or order, and the name of the contractor.

6. NOTES

6.1 **Ordering data.** — Purchasers should specify the class, color, and length of laces required and should exercise any desired options offered herein (see 1.2.1, 1.2.3, 3.2, 3.5, 4.2, 6.2, and 7.2.4.1).

6.2 It is believed that this specification adequately describes the characteristics necessary to secure the desired material and that normally no samples will be necessary prior to award to determine conformance to the requirements of this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.3 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.4 For informational purposes, 28/2 spun nylon is the yarn size accepted for meeting the physical requirements of this specification.

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6.5 Transportation description. — The transportation description shall be as follows:

Laces, shoe.
Carload weight 20,000 pounds, subject to Rule 34, Uniform Freight Classification.
Truckload weight 20,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standards, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, Marine Corps, and Air Force.

7.2.1 *Applicable specifications, standards, and other publication.*

Federal Specifications:
NN–B–601—Boxes, Wood-Cleated-Plywood, for Domestic Shipment.
NN–B–621—Boxes; Wood, Nailed and Lock-Corner.
UU–P–268—Paper, Kraft, Wrapping.

Military Specifications:
JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.
JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
MIL–L–10547—Liners, Case, Waterproof.

Navy Department Specification:
General Specifications for Inspection of Material.

Marine Corps Specification:
Boxes, Fiber, Corrugated — Standard, Half Standard, Quarter Standard.

Military Standards:
MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Standard sample.*—The laces shall be equal to the standard sample in all respects not specifically covered by this specification (see 7.2.6.2).

7.2.3 *Preproduction sample approval.*

7.2.3.1 *Army and Air Force.* — Before production is commenced, the contractor shall submit for approval to the contracting officer, six samples of the laces he proposes to furnish.

7.2.3.2 *Navy.*—The preproduction sample approval shall be as specified (see 7.2.6.1, item b).

7.2.3.3 *Marine Corps.*—Before production is commenced, six samples of the laces shall be submitted to the Depot Quartermaster, Marine Corps Depot of Supplies, 1100 South Broad Street, Philadelphia 46, Pa., for approval.

7.2.4 *Sampling and inspection.*

7.2.4.1 *Sampling.* — The sampling proce-

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dure shall be based on Military Standard MIL–STD–105, the Acceptable Quality Level to be as specified in the contract or order (see 7.2.6.1, item c); or the sampling procedure shall be as otherwise specified in the contract or order.

7.2.4.2 *Inspection.* — Inspection shall be made to determine compliance with the requirements of this specification.

7.2.4.2.1 *Visual inspection (Marine Corps).*—Visual inspection of the laces and the packaging, packing, and marking of the laces covered by this specification may be performed by Government inspectors in accordance with this specification and Military Standard MIL–STD–105. Acceptable Quality Levels shall be as specified.

7.2.4.2.2 *Inspection procedures (Navy and Marine Corps).* — For Naval and Marine Corps purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

7.2.5 *Preparation for delivery.*

7.2.5.1 *Packaging.* — One hundred forty-four laces, of one class and length only, shall be tied together and five such lots put up in a bundle. Each bundle shall be completely enclosed in a wrap of kraft paper having a minimum basis weight of 60 pounds per ream (24 x 36 — 500) conforming to Federal Specification UU–P–268. The wrap shall be sealed with 2-inch minimum width gummed kraft paper tape. Approximate dimensions of bundles shall be as specified in table II.

7.2.5.2 *Packing.*

7.2.5.2.1 *Army and Air Force.*—Laces of one class and length only, packaged as specified in 7.2.5.1 shall be packed in a shipping container for domestic and overseas shipment as hereinafter specified. Quantity and arrangement within the shipping containers shall be as specified in table II.

TABLE II.—*Packing details*

Class	Length	Over-all dimensions	Bundle				Shipping container		
			Arrangement in container				Inside dimensions approximate inches		
	Inches	*Inches*	*Quantity*	*Length*	*Width*	*Depth*	*Length*	*Width*	*Depth*
1	25	12½ x 7½ x 3½	16	2	2	4	27	15	14¼
1	27	13½ x 7½ x 3½	16	2	2	4	27	15	14¼
1	28½	14½ x 7½ x 3½	16	2	2	4	30	15	14¼
1	30	15 x 7½ x 3½	16	2	2	4	30	15	14¼
1	38	9½ x 15 x 3½	16	2	2	4	30	20	14¼
1	40	10 x 15 x 3½	16	2	2	4	30	20	14¼
1	50	25 x 7½ x 3½	8	1	2	4	27	15	14¼
1	52	26 x 7½ x 3½	8	1	2	4	27	15	14¼
1	54	27 x 7½ x 3½	8	1	2	4	27	15	14¼
1	84	21 x 15 x 3½	8	2	1	4	30	21	14¼
2	30	15 x 7½ x 7	8	2	2	2	30	15	14¼
2	40	10 x 15 x 7	8	2	2	2	30	20	14¼
2	44	11 x 15 x 7	8	2	2	2	30	22	14¼
2	52	26 x 7½ x 7	4	1	2	2	27	15	14¼
2	60	30 x 7½ x 7	4	1	2	2	30	15	14¼
2	64	16 x 15 x 7	4	2	1	2	30	16	14¼
2	72	18 x 15 x 7	4	2	1	2	30	18	14¼
2	80	20 x 15 x 7	4	2	1	2	30	20	14¼

NOTE: Bundle and container dimensions are approximations only and must be verified by contractor.

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7.2.5.2.2 *Navy.*—Shipments shall be made in accordance with Military Standard MIL–STD–129.

7.2.5.2.3 *Marine Corps.* — Four bundles (2,880 laces or 1,440 pairs) of class 1 laces shall be packed to a quarter standard, and class 2 laces to a standard size shipping container, conforming to Marine Corps Specification for Boxes, Fiber, Corrugated—Standard, Half Standard, Quarter Standard. The containers for the class 2 laces shall be lined with a one-piece unjoined liner made of the same material as the shipping container. Outside dimensions of the liner shall be the same as the inside dimensions of the container with corrugations to run in a vertical direction. Bottom flaps for all containers shall be securely stitched with 0.023-inch rust-resisting stitching wire. Top flaps shall not be glued or stitched, but securely held together with No. 1 kraft gummed paper tape. In the event this is insufficient to insure acceptance by carrier for safe transportation to destination, the containers shall be strapped with flat steel strapping. Bottom of all containers shall be lined with paper suitable for the purpose.

7.2.5.2.4 *Shipping containers (Army and Air Force).*

7.2.5.2.4.1 *For domestic shipment.*—Shipping containers shall conform to style A of Federal Specification NN–B–601, or style 4 of Federal Specification NN–B–621.

7.2.5.2.4.2 *For overseas shipment.*—Shipping containers shall be constructed and strapped in accordance with style A of Military Specification JAN–P–105, or style 4 of Military Specification JAN–P–106. Each overseas shipping container shall be provided with a type I or II, grade B, class 2, L-2 or M case liner, in conformance with Military Specification MIL–L–10547.

7.2.5.3 *Marking.*

7.2.5.3.1 *Packages.*

7.2.5.3.1.1 *Army and Air Force.* — Each bundle shall be legibly marked with the following information printed on the label:

Nomenclature
Class and length
Quantity
Stock number
Specification number
Contract number and date
Name of contractor
Contracting bureau or agency

7.2.5.3.1.2 *Marine Corps.* — In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Military Standard MIL–STD–129.

7.2.5.3.2 *Shipping containers.*

7.2.5.3.2.1 *Army and Air Force.* — Each shipping container shall be marked in accordance with Military Standard MIL–STD–129 except that corner service color markings shall not be applied to containers packed for domestic shipment. The nomenclature shall be: Laces, Footwear, Spun Nylon.

Shape *_____
Length *_____
Color *_____
Fed. Spec. V–L–61a

7.2.5.3.2.2 *Marine Corps.*—Each shipping container shall be marked in accordance with Military Standard MIL–STD–129.

7.2.6 *Notes.*

7.2.6.1 *Ordering data.*—Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Preproduction sample approval fixture (see 7.2.3.2).

(*c*) Acceptable quality level (see 7.2.4.2.1).

* Applicable data to be entered by contractor.

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(*d*) Whether packing for domestic or overseas shipment is required (see 7.2.5.2).

7.2.6.2 *Standard sample.*

7.2.6.2.1 *Army and Air Force.*—For access to standard sample, address the procuring office issuing the invitation for bids.

7.2.6.2.2 *Navy.*—The standard sample for Bureau of Supplies and Accounts procurements may be obtained upon application to the Clothing Supply Officer, Clothing Supply Office, Twenty-ninth Street and Third Avenue, Brooklyn 32, N. Y.

7.2.6.2.3 *Marine Corps.*—Standard sample may be obtained upon application to the Depot Quartermaster, Marine Corps Depot of Supplies, 1100 South Broad Street, Philadelphia 46, Pa.

7.2.6.3 The following list of footwear requires the lace length specified opposite the nomenclature:

	Inches
Shoe, boat, safety sole	25
Shoe, women's low quarter	27
Shoe, nurse's white	27
Shoe, low quarter, tan (sizes 5 to 12)	28½
(size 12½ and up)	30
Shoe, field, women's	38
Boot, service, women's	40
Shoe, safety toe	40
Boot, leather, legging top	40
Boot, combat, rubber, insulated	60
Boot, combat, tropical	40
Boot, arctic, felt	44
Boot, mukluk	30 or 60
Boot, mukluk, cold-dry	34 or 44
Boot, ski-mountain	60
Boot, service, combat, russet	64
Shoe, convalescent	52 or 84
Shoepac	72
Boot, blucher, high top	80
Overshoe, rubber, women's high, laced fastening	50

NAVY INTEREST: *MC* S

ARMY INTEREST: Q

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